Health Through Evidence

Real-World Data, Secondary Use and AI in the Hospital

From connected systems
to decisions that improve care.



George Gourzoulidis, PhD

Managing Partner & Co-Founder

Health Through Evidence

The infrastructure is the start.
The decision is the point.

01	ACCESS	**EHDS**	Trusted secondary use
02	CONNECTION	**Interoperability**	EHR and hospital prescribing
03	DECISION	**Hospital evidence**	What worked, for whom, at what cost?

Secondary use: re-use data collected during care to answer a new, defined question.

Digital does not yet mean decision-ready.

WHAT EXISTS, WHAT IS MISSING

SOURCES TO ASSESS

Prescribing & pharmacy

Exposure, dispensing, administration*

EHR & disease registries

Clinical context and recorded outcomes

DRG activity & accounting

Episodes, resource use and local costs

Patient-reported measures

Symptoms, function and quality of life

WHAT STILL NEEDS WORK

LINK	Match people, episodes and time.
DEFINE	Use consistent endpoints and costs.
VALIDATE	Check missingness, bias and quality.
GOVERN	Name owners; approve secure access.

Rollout is not proof of complete, linkable data.

Sources: national EHR (2025); hospital e-prescribing decision (28 Aug 2026); NICE RWE framework.

Follow the patient.
Not the information system.



One linked cohort. One common time horizon. A defined outcome.

Conceptual patient pathway; source-selection principles: NICE RWE framework.

Greek prescribing data: use, uptake, cost.



Current Oncology

MDPI

Article

Real-World Prevalence, Treatment Patterns, and Economic Impact of EGFR- and ALK-Targeted Therapies in Non-Small Cell Lung Cancer: A Nationwide Analysis from Greece

George Gourzoulidis [1,*], Catherine Kastanioti [1], George Mavridoglou [2], Theodore Kotsilieris [1], Anastasios Tsolakidis [3], Konstantinos Mathioudakis [3], Dikaios Voudigaris [4] and Charalampos Tzanetakos [4]

[1] Department of Business and Organizations Administration, University of the Peloponnese, 24100 Kalamata, Greece; a.kastanioti@uop.gr (C.K.); t.kotsilieris@uop.gr (T.K.)
[2] Department of Accounting and Finance, School of Management, University of the Peloponnese, 24100 Kalamata, Greece; ge.mavridoglou@go.uop.gr
[3] IDIKA SA—E-Government Center for Social Security Services, 10551 Athens, Greece; tsolakidis@idika.gr (A.T.); mathioudakis@idika.gr (K.M.)
[4] Health Through Evidence GP, 17456 Athens, Greece; d.voudigaris@hte.gr (D.V.); c.tzanetakos@hte.gr (C.T.)
* Correspondence: g.gourzoulidis@go.uop.gr or gourzoulidis.g@gmail.com

Simple Summary

Lung cancer is the leading cause of cancer-related deaths in Greece and worldwide. In recent years, new treatments called tyrosine kinase inhibitors (TKIs) have been developed to target specific genetic mutations in lung cancer, such as EGFR and ALK. These medicines often work better and have fewer side effects than traditional chemotherapy. To understand how these treatments are used in Greece, we analyzed nationwide prescription data from 2020 to 2022. We looked at all patients who began treatment with an EGFR- or ALK-targeted drug during this period. Our study found that Greek clinical experts quickly switched from older TKIs to newer ones, such as osimertinib and alectinib, which are now the most commonly prescribed. Although the number of treated patients was relatively small, the cost to the national health system was still significant. However, total spending remained stable over time. This was largely due to national pricing regulations and negotiated discounts, which helped control costs despite the use of newer and more expensive drugs. These findings provide valuable insights for health authorities and policymakers, helping them ensure fair and sustainable access to innovative cancer treatments for patients in Greece.

check for updates

Received: 8 August 2025
Revised: 25 September 2025
Accepted: 26 September 2025
Published: 27 September 2025

Citation: Gourzoulidis, G.; Kastanioti, C.; Mavridoglou, G.; Kotsilieris, T.; Tsolakidis, A.; Mathioudakis, K.; Voudigaris, D.;

GREEK SECONDARY-USE STUDY

Secondary-use prescribing data can generate epidemiology, treatment-pattern and cost evidence.

Nationwide IDIKA e-prescribing | First-line EGFR/ALK-targeted NSCLC therapy | 2020–2022

EPIDEMIOLOGY

Prescribing-based prevalence and patient characteristics

TREATMENT PATTERNS

EGFR- and ALK-targeted therapy use and changes over time

COST

Annual drug expenditure and per-patient treatment costs

Existing data. New evidence for policy and HTA.

From national prescribing to hospital-level value.

Health Through Evidence

ALREADY DEMONSTRATED

National prescribing

Who was treated
Which medicine
Estimated drug spend

WHAT THE HOSPITAL COULD ADD

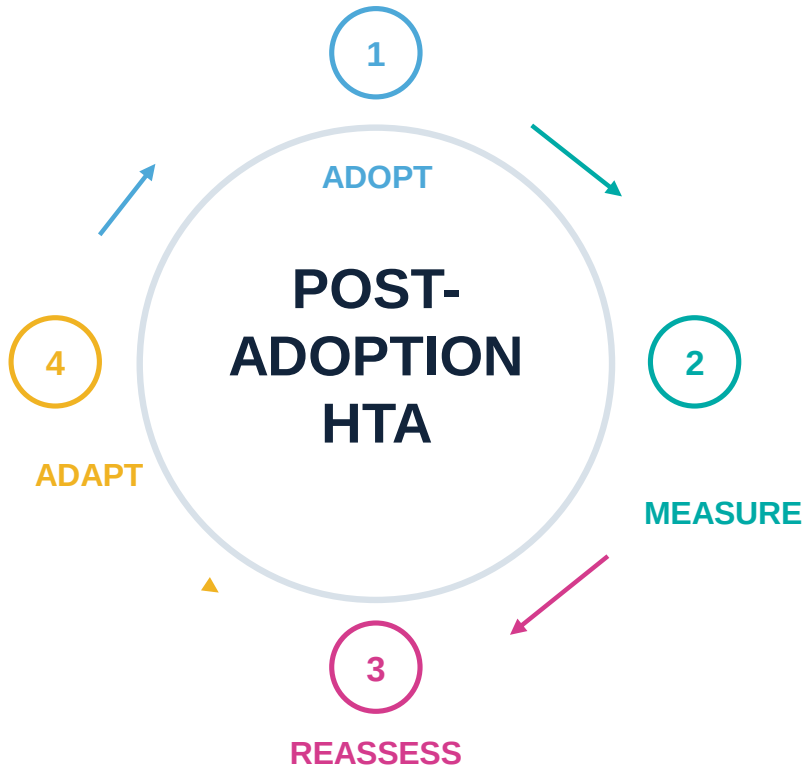
Confirmed cohort	Stage, biomarker, line of therapy
Actual treatment	Administration, dates, switching
Pathway evidence	Resources, outcomes, follow-up

METHOD GATE

Comparator + baseline risk + missing data + uncertainty

After adoption, value must be checked.

HTA AND OUTCOME-BASED AGREEMENTS



BEFORE PAYMENT FOLLOWS OUTCOMES

Eligible cohort	Who counts?
Endpoint + horizon	What result, measured when?
Baseline + missingness	How are risk and absent data handled?
Audit + consequence	Who verifies, and what changes?

The hospital supplies auditable evidence.
The authorised payer and contracting parties define the agreement.

NICE RWE framework; NICE managed access. Evidence requirements for outcome-linked agreements.

An outcome can trigger a payment.

Yescarta (axicabtagene ciloleucel) | Italy | Documented launch agreement, 2019

DRUG PAYMENT AT INFUSION

None

OUTCOME-CONDITIONAL INSTALMENTS



Payment was conditional on a positive treatment outcome.

HOSPITAL ROLE	Registry follow-up → outcome documentation → hospital payment outside the registry

Scheme closed to new patients on 14 December 2022. Follow-up continued for existing agreements.

Evidence for reassessment and payment by results are different mechanisms.

AIFA: Yescarta reimbursement announcement, 13 Nov 2019; registry/MEA closure notice, 15 Dec 2022.

Spending is not the same as value.

Connect the drug budget to the outcomes and resources of the same patient cohort.

BUDGET CONTROL

What did we spend?

Actual drug expenditure versus approved budget

Patients treated and therapy mix

WHAT MUST BE LINKED

PATHWAY COST	Medicines, administration, admissions, monitoring and adverse events
OBSERVED OUTCOMES	Effectiveness, safety and patient experience, with adequate follow-up
MANAGEMENT DECISION	**Review use. Plan capacity. Justify resources.**

Same cohort. Same horizon. Comparable case-mix.

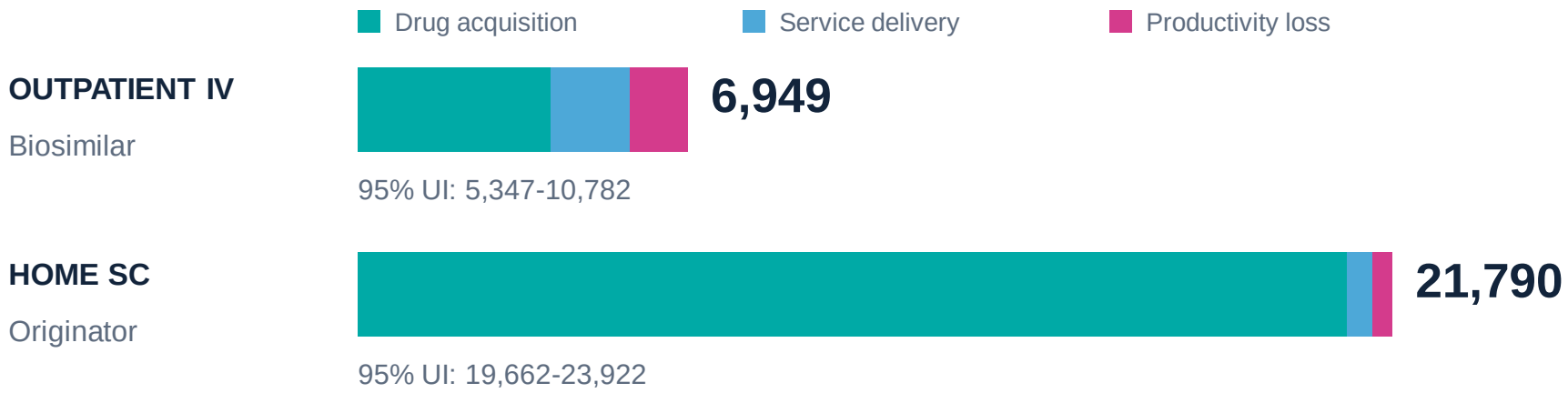
Sources: NICE technology appraisal methods, resource impact (4.11); NICE real-world evidence framework.

A cheaper workflow is not always a cheaper pathway.

FROM PATHWAY EVIDENCE TO A BUDGET DECISION

Singapore, 2026 publication | Trastuzumab: outpatient IV biosimilar versus home SC originator

ANNUAL SOCIETAL COST PER PATIENT | 2024 S$



-66%

service delivery cost

+214%

total societal cost

MANAGEMENT IMPLICATION

BUDGET

Use local net drug prices

CAPACITY

Verify usable staff and chair time

ADOPTION

Re-cost before scaling the pathway

Ouyang et al. Ther Adv Med Oncol. 2026;18. doi:10.1177/17588359261421557.

Where AI can support the hospital

Clinical support, everyday work and the generation of usable evidence.

CLINICAL SUPPORT

Imaging interpretation and deterioration risk alerts

Assess: clinical benefit, safety and timeliness

MASAI (2026); CONCERN (2025; corrected 2026)

CLINICAL DOCUMENTATION

Draft visit notes and summarise encounters

Assess: clinician time and record accuracy

Lukac et al., NEJM AI (2025)

FLOW AND CAPACITY

Anticipate admissions and support bed planning

Assess: waiting time and patient flow

Nat Commun (2026), doi:10.1038/s41467-026-72960-1

SECONDARY USE

Extract structured variables from clinical text

Assess: data quality and research efficiency

Yuan et al., Communications Medicine (2025)

Different applications. Different evidence. Local benefit must be demonstrated.

A real trial. A measurable workflow benefit.

MASAI | Randomised population screening trial | 105,915 women analysed

INTERVAL CANCER RATE

1.55 vs 1.76

per 1,000 participants: AI vs standard reading



Ratio 0.88 (95% CI 0.65-1.18)

Non-inferior; superiority was not established.

SENSITIVITY

80.5% vs 73.8%

Higher with AI (p=0.031)

SPECIFICITY

98.5% in both groups

44%

less screen-reading workload*

Gommers et al. Lancet 2026;407:505-514. *Reading workload: earlier MASAI safety analysis, Lang et al. 2023.

Buying AI starts an evaluation. It does not finish one

01

DEFINE

Intended use
and clinical owner

What decision will it change?

02

VALIDATE

Local data, subgroups
and calibration

Does it work here?

03

PILOT

Patient benefit,
workload
and unintended harm

Is the net effect worthwhile?

04

MONITOR

Drift, incidents
and a stop rule

Who can pause or override?

Human oversight / Privacy and security / Version and incident logs / Documented accountability

European Commission AI Act overview (updated 19 Oct 2026); NICE RWE principles. Proposed hospital adoption checks.

Be ready to describe, govern and supply.

EHDS: THE HOSPITAL AS DATA HOLDER

2025	2027	2029	2031
In force	Key implementing acts	Most secondary-use rules	Remaining categories

HOSPITAL / DATA HOLDER

Describe datasets
Maintain metadata annually
Document quality
Prepare requested data

HEALTH DATA ACCESS BODY

Assess applications
Authorise lawful purposes
Coordinate access
Apply safeguards

SECURE ANALYSIS

Access-controlled environment
No re-identification
Output disclosure checks
Protect individual rights

The hospital prepares data. It does not independently authorise unrestricted reuse.

Regulation (EU) 2025/327, Art. 60, 68, 73 and 77; European Commission EHDS timeline and HealthData@EU.

Make evidence a management function.

The pilot starts the work. Governance sustains it.

OWNER	Executive sponsor + data steward
CATALOGUE	Datasets, quality, access rules
ANALYSIS	Priority questions + explicit methods
DECISION	Quarterly brief, action, follow-up

FIRST 90 DAYS

1	**accountable owner**
10	**priority datasets mapped**
3	**management questions**
1	**decision brief**

One pathway. One minimum dataset. Twelve months.

Proposed operating model and first-90-day targets; adapt scope and resources to local readiness.

Thank you!

Health Through Evidence

Contact us:

Email: g.gourzoulidis@hte.gr

CMT Prooptiki
CONSULTING MANAGEMENT TRAINING